

LECTURE II

QUANTUM QUERY COMPLEXITY LOWER
BOUNDS

All quantum algos we saw are in the
blackbox query model.

Given input $F: \{0, 1\}^n \rightarrow \{0, 1\}^m$

or $F: \{0, 1, \dots, N-1\} \rightarrow \{0, 1\}^m$

via quantum circuit Q_F^*

$N = 2^n$

Solve some problem ϕ about F .

Examples

(1) Simon's problem

$$\forall x \in \{0,1\}^n$$

given $F(x) = F(x+L)$ for some
 $L \in \{0,1\}^n \setminus \{0^n\}$

Find L .

(2) Grover's problem:

given $F : \{0,1\}^n \rightarrow \{0,1\}$

Find x s.t. $F(x) = 1$

Our algorithms never "looked inside" Q_F^* .

Query complexity model:

- Q_F^* is a black box
- # of queries = # of times we apply Q_F
- All other computation is "free"

Natural Analogues: Deterministic/Randomized

Why study this model?

- All our quantum algs fit this model
- The computational complexity we ignore/don't account for is usually small
- The model is **simple enough** that it allows us to prove lower bounds
Clean characterization of Algorithm $\leftarrow$ Input interaction
- Gives evidence about power of randomness/quantum

We will slightly simplify notation

Instead of querying $F: \{0, 1, \dots, N-1\} \rightarrow \{0, 1\}^m$

We will assume we query $\{w_0, w_1, \dots, w_{N-1}\}$

$w_x := F(x)$, an m-bit string

$$\underline{Q_F^*} : \underset{\substack{\uparrow \\ n \text{ qubits}}}{|x\rangle} \otimes \underset{\substack{\uparrow \\ m \text{ qubit}}}{|b\rangle} \rightarrow \underline{|x\rangle} \otimes \underline{|b \oplus w_x\rangle}$$

Our goal : Decision tasks [for simplicity]

Comput $\phi : \{w_0, \dots, w_{N-1}\} \rightarrow \{0,1\}$

Example : Decision Grover

$w_i \in \{0,1\}$

Goal : Does there exist x s.t. $w_x = 1$

$$\phi(w_0, \dots, w_{N-1}) = w_0 \vee w_1 \vee w_2 \vee \dots \vee w_{N-1}$$

Search Grover is at least as hard as decision Grover

- Run search Grover
- Get x
- Check if $f(x) = 1$ if True, output YES
else output NO

Decision Simon

$$F(x) = F(x \oplus s)$$

$$w_i \in \{0,1\}^m$$

Promise : $\exists s \in \{0,1\}^n$ s.t. $\forall i,j$

$$w_i = w_j \quad \text{if and only if} \quad \text{base2}(i) \oplus \text{base2}(j) = \underline{s}$$

Decide : Is $s = 0^n$ (i.e. the function is 1-1)

or $s \neq 0^n$ (i.e. the function is periodic with period s)

Total problems vs Promise problems

Total problem: All input $w_0, \dots, w_{N-1}$ are valid

Promise problem: Input is restricted, satisfies a promise

Let ϕ be a decision problem

$D(\phi)$: least # of queries needed by a deterministic algo to compute $\phi(w)$ for all w .

$R(\phi)$: least # of queries by a randomized algo that $\forall w$ computes $\phi(w)$ w.p. $\geq 2/3$

$Q(\phi)$: least # of queries by a quantum algo that $\forall w$ computes $\phi(w)$ w.p. $\geq 2/3$

Example : $\phi = \text{OR}$ [Grover's Problem]

$$D(\phi) = N$$

$$R(\phi) = \left\lceil \frac{2}{3} N \right\rceil$$

$$Q(\phi) \leq \sqrt{N}$$

We will see soon that $Q(\phi) \geq \sqrt{N}$

Example : Decision Simon

$$D(\emptyset) = \Theta(\sqrt{N}) \leftarrow \text{Nice Puzzle}$$

$$R(\emptyset) = \Theta(\sqrt{N})$$

$$Q(\emptyset) = \log N$$

$F(0 \dots 0 \text{ blal}) \rightarrow \sqrt{N}$
outputs

$F(\text{blal } 0 \dots 0)$
 $\downarrow$
 $\sqrt{N}$ outputs

Collision exists iff period
is non-zero

Example: 2-to-1 problem

Promise : Either all w_i are distinct
or each w_i repeated *exactly twice*

$$\left. \begin{array}{l} D(\phi) = \frac{N}{2} + 1 \\ R(\phi) = N^{1/2} \end{array} \right\} \begin{array}{l} Q(\phi) \leq N^{1/3} \\ \text{Modify Grover Search (non-trivial)} \end{array}$$

Example: Element Distinctness

Does w have repetitions or all w_i 's distinct

$$D(\emptyset) = N$$

$$R(\emptyset) = N$$

$$Q(\emptyset) = \underline{\theta(N^2)}$$

Example: Recursive Majority tree

$$w = \begin{array}{ccc} \underline{110} & 010 & 011 \\ \swarrow \downarrow \searrow & \swarrow \downarrow \searrow & \swarrow \downarrow \searrow \end{array}$$

$$N = \underline{3^h}$$



height h

$$D(\emptyset) = 3^h = N$$

$$R(\emptyset) = \left(\frac{8}{3}\right)^h \approx \underline{N^{0.887}}$$

$$Q(\emptyset) = 2^h \approx N^{0.63} / \text{variant of Grover search}$$

This is a
total problem

Total Problems vs Promise Problems

For total problems, there is only a polynomial gap between deterministic, randomized and quantum

Exponential gap only for promise problems

This is not a coincidence

Theorem [BBCMW 98]

$\forall$ total $\phi: \{0,1\}^N \rightarrow \{0,1\}$

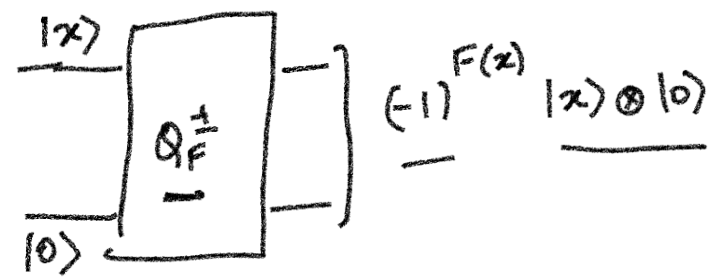
$$D(\phi) \leq Q(\phi)^6 \quad \text{i.e.} \quad Q(\phi) \geq (D(\phi))^{1/6}$$

Theorem [Nisan 89]

$$\text{For total } \phi, \quad D(\phi) \leq R(\phi)^3 \quad \text{i.e.} \quad R(\phi) \geq (D(\phi))^{1/3}$$

Quantum lower bound for OR function
(Grover search)

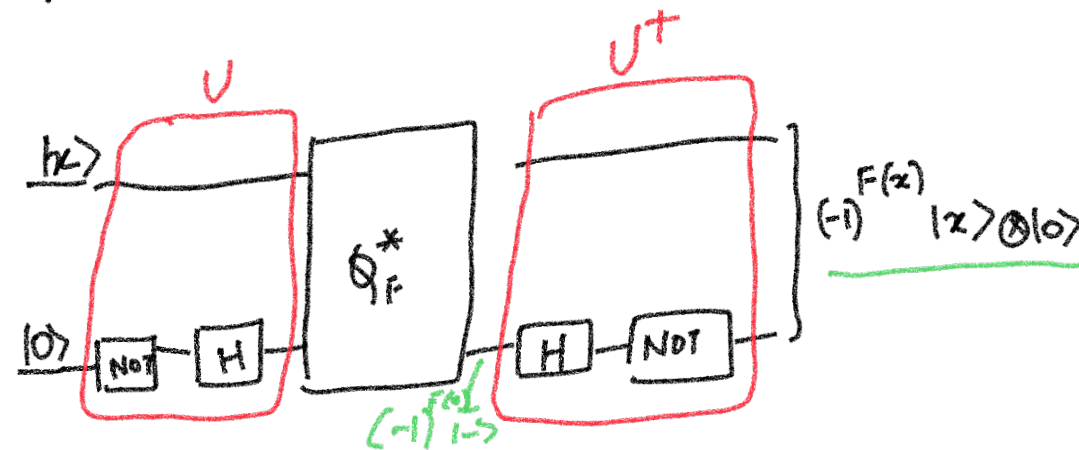
Equivalence of Q_F^* and $Q_F^\pm$



$$F: \{0, 1\}^n \rightarrow \{0, 1\}$$

$$\sum \alpha_i |x\rangle \otimes |0\rangle \xrightarrow{Q_F^{-1}}$$

$$\sum \alpha_i (-1)^{F(x)} |x\rangle \otimes |0\rangle$$



$Q_F^\pm$ is the same as U Q_F^* $U^\dagger$

equivalently

Q_F^* is $U^\dagger Q_F^\pm U$

$$\underline{U_0} \underline{Q_F^*} \boxed{U_1} \underline{Q_F^*} \underline{U_2} \underline{Q_F^*} \dots$$

is equivalent to

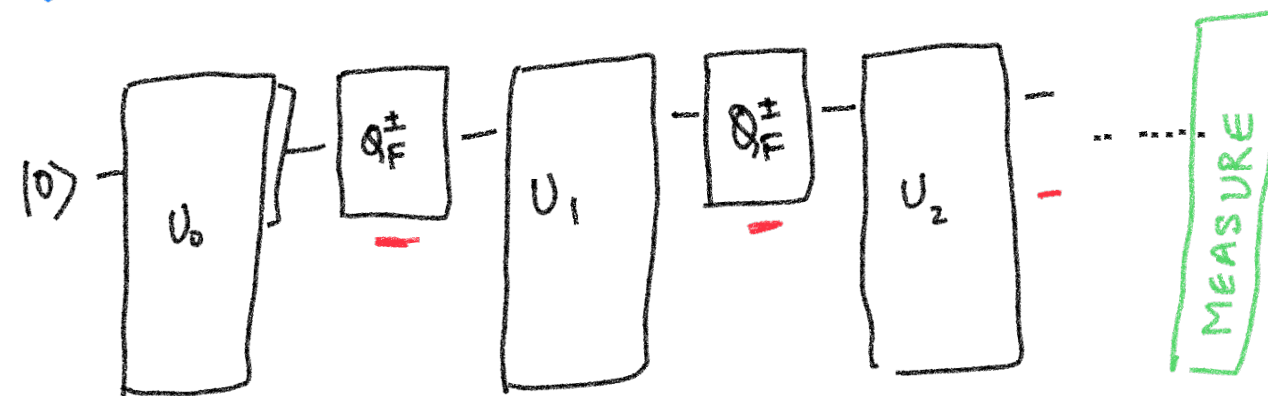
$$\underline{U_0} \underline{U^+ Q_F^\pm} \underline{U_1} \underline{U^+ Q_F^\pm} \underline{U_2} \underline{U^+ Q_F^\pm} \dots$$

$$U'_0 Q_F^\pm \quad U'_1 Q_F^\pm \quad U'_2 Q_F^\pm \dots$$

w.l.o.g. we assume queries to w are always $Q_F^\pm$

We want to look at minimum number of queries
to compute $\phi(w) = w_0 \vee w_1 \vee \dots \vee w_{N-1}$

Consider an arbitrary circuit computing ϕ



Why does a circuit look like this w.l.o.g.?

Best chance of distinguishing between

$$\frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \quad |\phi\rangle$$

and

$$\frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ -1 \\ \vdots \\ 1 \end{bmatrix} \quad |\psi\rangle$$

Best distinguishing advantage is

$$1 - |\langle \phi | \psi \rangle|^2$$
$$1 - \underbrace{\left(1 - \frac{2}{N}\right)}_{\approx 0} = \frac{2}{N}$$

Recall : How do we distinguish two states

$|u\rangle$ and $|v\rangle$?

What is the best algorithm?

Measure in $|u\rangle, |u^\perp\rangle$

given $|u\rangle$, we output $|u\rangle$
w.p. 1

given $|v\rangle$ we output $|u\rangle$
w.p. $|\langle v|u\rangle|^2$

Measure in $|v\rangle, |v^\perp\rangle$

We distinguish with adv.
 $1 - |\langle v|u\rangle|^2$

Question: does this mean we cannot
distinguish

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

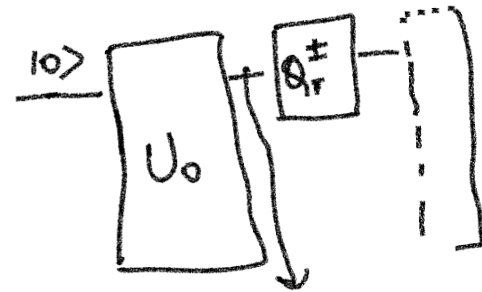
and

$$\begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$Q_F^\pm$$

YES : $w \in \{0,1\}^N \setminus 0^N$

NO : 0^N



YES input and NO input look identical

For YES case : $0 \dots 0 \overset{1}{\underset{\substack{\text{ } \\ \text{ } \rightarrow j\text{-th position is 1}}}{0}} \dots 0$

$$|\phi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle + \dots + \alpha_{N-1}|N-1\rangle \otimes |y\rangle$$

$$\downarrow Q_F^\pm$$

$$\alpha_0|0\rangle + \alpha_1|1\rangle + \dots + \alpha_{j-1}|j-1\rangle - \alpha_j|j-1\rangle + \alpha_{j+1}|j+1\rangle + \dots$$

$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle + \dots + \alpha_{N-1}|N-1\rangle \otimes |y\rangle$$

$$\downarrow Q_F^\pm$$

$$\alpha_0|0\rangle + \alpha_1|1\rangle + \dots + \alpha_{j-1}|j-1\rangle + \dots + \alpha_{N-1}|N-1\rangle$$

$$|\langle \emptyset | \psi \rangle| = |\alpha_0|^2 + \dots + |\alpha_{N-1}|^2 - 2|\alpha_j|^2$$

$$= 1 - 2|\alpha_j|^2$$

To distinguish, we want $|\alpha_j|$ to be large.

If we know j then we can choose U_0 st. $|\alpha_j|$ is large and we are able to distinguish

Distinguishing between

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

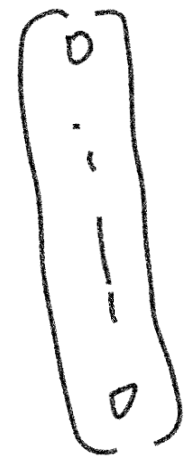
and

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

is

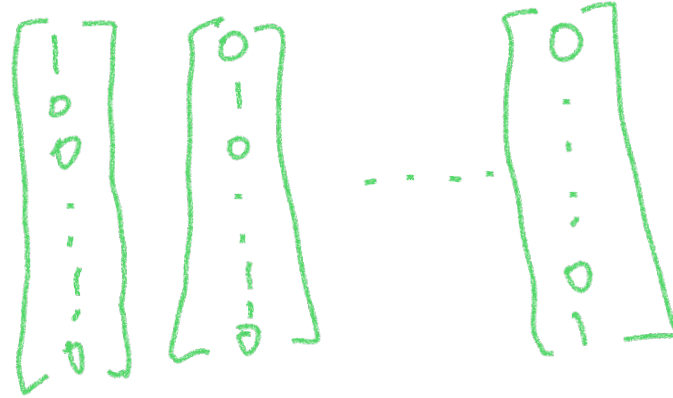
easy.

We want to prove that distinguishing between
the following is hard



NO

One of the following



YES

$|\phi_0\rangle, \dots, |\phi_{N-1}\rangle$ are possible YES instances

$|\psi\rangle$ is a NO instance

After i query operations of the quantum query algorithm let the states be

$|\phi_0^{(i)}\rangle, \dots, |\phi_{N-1}^{(i)}\rangle$

$|\psi^{(i)}\rangle$

$$\sum_{i=0}^{N-1} K \psi^{(0)} | \phi_i^{(0)} \rangle = N$$

Suppose the algorithm succeeds in distinguishing after t operations.

$$\sum_{i=0}^{N-1} K \psi^{(t)} | \phi_i^{(t)} \rangle < 0.9 N$$

Claim $\forall i$

$$\sum_{i=0}^{N-1} |\langle \psi^{(i)} | \phi_i^{(j)} \rangle| - \sum_{i=0}^{N-1} |\langle \psi^{(i-1)} | \phi_i^{(i-1)} \rangle|$$

is "small"

This will imply t must be "large"

Fix i ,

$$|\psi^{(i-1)}\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle + \dots + \alpha_{N-1} |N-1\rangle \otimes \dots$$

$$|\phi_i^{(i-1)}\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle + \dots + \beta_{N-1} |N-1\rangle \otimes \dots$$

After $Q_F^\pm$

$$|\psi^i\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle + \dots + \alpha_{N-1} |N-1\rangle \otimes \dots$$

$$|\phi_i^i\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle + \dots + \beta_{i-1} |i-1\rangle - \beta_i |i\rangle + \beta_{i+1} |i+1\rangle + \dots \otimes \dots$$

$$\begin{aligned}
& \sum_{i=0}^{N-1} |\langle \psi^{(j)} | \phi_i^{(j)} \rangle| - |\langle \psi^{(j-1)} | \phi_i^{(j-1)} \rangle| \\
& \leq \sum_{i=0}^{N-1} 2 |\alpha_i| |\beta_i| \\
& \leq 2 \sum_{i=0}^{N-1} |\alpha_i| \leq 2\sqrt{N}
\end{aligned}$$

Recall $\sum_{i=0}^{N-1} |\alpha_i|^2 = 1$

$$\Rightarrow \sum_{i=0}^{N-1} |\alpha_i| \leq \sqrt{N} \quad \left(\begin{array}{l} \text{Cauchy Schwarz} \\ \text{or} \\ \text{A.M. - R.M.S. ineq.} \end{array} \right)$$

$$N - 0.9N$$

$$\leq \sum_{i=0}^{N-1} (\langle \phi^{(t)} | \psi_i^{(t)} \rangle - \langle \phi^{(0)} | \psi_i^{(0)} \rangle)$$

$$\leq 2\sqrt{N} \cdot t$$

$$t \geq \frac{0.1N}{2\sqrt{N}} = 0.05\sqrt{N}$$